

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. The transition action contains only dynamic contact entropy; static preparation is charged once. Resolvent-smoothed cylinders form a genuine graph core, and correctors solve the actual finite augmented generator equation. A two-scale slope-capped penalty localizes doubled states by sending its slope bound to infinity, giving comparison and nonlinear semigroup convergence.

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CONTENTS

1. Round-eight reconstruction: dynamic action, resolvent graph core, and two-scale comparison	1
1.1. Dynamic-only transition action	2
1.2. A resolvent-generated graph core	2
1.3. Exact finite-generator correctors	3
1.4. A two-scale slope-capped penalty	3
1.5. Nonlinear semigroup convergence	4
Scope and verification status	4

1. ROUND-EIGHT RECONSTRUCTION: DYNAMIC ACTION, RESOLVENT GRAPH CORE, AND TWO-SCALE COMPARISON

Static preparation is charged once in the initial boundary functional. The transition action contains only dynamic contact entropy. Smooth graph-core approximation uses resolvents of the actual generator, not an impossible analytic cutoff. Comparison uses a slope-capped penalty whose coercivity is sent to infinity in a separate limit.

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1.1. Dynamic-only transition action. For a compatible path (f, Γ) on $[s, t]$ set

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f dr, \quad \ell(q) = q \log q - q + 1.$$

The preparation rate $I_0(f_s)$ is not part of $\mathcal{A}_{s,t}$.

Lemma 1.1 (Exact cocycle of dynamic cost). *For every compatible path and $s < u < t$,*

$$\mathcal{A}_{s,t} = \mathcal{A}_{s,u} + \mathcal{A}_{u,t}.$$

Consequently the Lax–Oleinik operators

$$(V_t \varphi)(f) = \sup \{ \varphi(f_t) - \mathcal{A}_{0,t} : f_0 = f \}$$

satisfy $V_{t+s} = V_t V_s$.

Proof. The action is an integral over time, so additivity is exact. Concatenation of compatible paths preserves the B2 graph trace at the joining time. Splitting and concatenating competitors proves both semigroup inequalities. A static rate can be applied once to an initial distribution, but is never recharged at the intermediate state. $\square$

1.2. A resolvent-generated graph core. Let $\mathbb{A}_\varepsilon$ be the closed augmented BBGKY generator on the law/contact-ledger state and

$$J_{\lambda,\varepsilon} = (I - \lambda \mathbb{A}_\varepsilon)^{-1}.$$

For a finite coordinate projection Π_m and $\phi \in C_c^\infty(\mathbb{R}^m)$ define

$$F_{m,\lambda}(z) = \phi(\Pi_m J_{\lambda,\varepsilon} z).$$

Lemma 1.2 (Smooth bounded graph core). *The algebra generated by $F_{m,\lambda}$ lies in the generator graph domain, is stable under products, and is dense on every action compactum in the C_b^2 cylinder graph norm as $m \rightarrow \infty$ and $\lambda \downarrow 0$.*

Proof. The resolvent maps the state space into $D(\mathbb{A}_\varepsilon)$ and satisfies $\mathbb{A}_\varepsilon J_\lambda = \lambda^{-1}(J_\lambda - I)$. The chain rule gives bounded graph images because ϕ has compactly supported derivatives. The Markov generator product formula on its resolvent range gives stability under products. Compact containment makes finitely many coordinates uniformly separating; Stone–Weierstrass approximates the cylinder, and $J_\lambda \rightarrow I$ uniformly on the compactum. No analytic function is required to equal one on an open box and change outside it. $\square$

1.3. Exact finite-generator correctors. Let $V_\varepsilon(t, s)$ be the backward observable propagator of the actual finite augmented generator along a smooth reference path:

$$\partial_t V_\varepsilon(t, s) = -\mathbb{A}_\varepsilon(t) V_\varepsilon(t, s), \quad V_\varepsilon(s, s) = I.$$

For the order- j defect D_j^ε define

$$c_j^\varepsilon(t) = \int_t^T V_\varepsilon(t, s) D_j^\varepsilon(s) ds.$$

Theorem 1.3 (Finite graph-domain Poisson corrector). *Each corrector belongs to the finite generator graph domain and satisfies*

$$\partial_t c_j^\varepsilon + \mathbb{A}_\varepsilon(t) c_j^\varepsilon = -D_j^\varepsilon, \quad c_j^\varepsilon(T) = 0.$$

For suitable ledger weights,

$$\sum_{j \geq 0} \sup_{t \leq T} (\|c_j^\varepsilon(t)\| + \|\mathbb{A}_\varepsilon c_j^\varepsilon(t)\|) < \infty$$

uniformly in ε .

Proof. The propagator is constructed by forward Picard iteration in remaining time and preserves the graph domain by the B2 boundary-renewal estimate. Differentiating the Duhamel integral gives the equation and the exact zero terminal value. A defect with j ledger labels carries the cluster factor $C^j/j!$; choose the ledger radius below the graph-semigroup radius and sum the resulting exponential series, including the generator images. $\square$

1.4. A two-scale slope-capped penalty. Let d_B be a bounded-compatible metric on an energy containment set. Choose $\omega \in C^2([0, \infty))$ increasing, $\omega(r) = r$ for $r \leq 1$, and $0 \leq \omega' \leq 1$. For $R \geq 1$ and $\eta > 0$ put

$$\Psi_{R,\eta}(f, g) = \frac{R}{\eta} \omega(\eta d_B(f, g)).$$

For fixed R , its cotangent norm is at most R . Coercivity is obtained by $R \rightarrow \infty$, not by falsely claiming that $\eta \downarrow 0$ makes a bounded distance diverge.

Lemma 1.4 (Diagonal localization with bounded slopes). *At a doubled maximum of $u(f) - v(g) - \Psi_{R,\eta}(f, g)$ on a common containment set,*

$$d_B(f, g) \leq C/R + o_\eta(1),$$

while both test cotangents lie in the Hamiltonian source ball of radius R .

Proof. Compare the maximum with the diagonal point $f = g$. Since u, v are bounded, $\Psi_{R,\eta}(f, g) \leq C$. For $\eta d_B \leq 1$ the penalty is Rd_B ; if this inequality fails, first choose η small on the bounded metric space. Thus $d_B \leq C/R$. Differentiation gives slope $R\omega'(\eta d_B) \leq R$. $\square$

Theorem 1.5 (Comparison on the full state space). *Assume the Hamiltonian is continuous on every bounded cotangent ball and obeys the exponential containment bound supplied by B2. Then bounded upper and lower viscosity solutions of the Hamilton–Jacobi equation satisfy comparison.*

Proof. Fix R and apply the doubled-variable argument with $\Psi_{R,\eta}$. Lemma 1.4 keeps the two cotangents inside the radius- R chart. Let $\eta \downarrow 0$ to remove smoothing of the metric. The states are then within C/R . Hamiltonian continuity on the radius- R chart makes the state mismatch vanish as $R \rightarrow \infty$; exponential containment removes the complement before this limit. The standard viscosity inequalities then contradict a positive maximum. $\square$

1.5. Nonlinear semigroup convergence.

Theorem 1.6 (Microscopic-to-kinetic nonlinear semigroups). *The logarithmic microscopic semigroups on the augmented law/contact-ledger state converge locally uniformly on action compacta to the dynamic Lax–Oleinik semigroup generated by $\mathcal{A}$. The limit is unique and satisfies the semigroup law.*

Proof. Use the graph core of Lemma 1.2 and the exact correctors of Theorem 1.3 to obtain generator convergence with no terminal remainder. B2 gives compact containment and lower recovery for the dynamic action. Half-relaxed limits are viscosity sub- and supersolutions. The comparison theorem identifies them. Lemma 1.1 supplies the semigroup law and prevents double charging of preparation. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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